

WIRING UP THE VALUE CHAIN

From smartphones and electronic components to semiconductors, India is building the deeper supply chains needed to turn manufacturing scale into global competitiveness

BY PALAK AGARWAL

198 |

▶ **INDIA'S ELECTRONICS MANUFACTURING** story is entering a new phase. After smartphones, which became the symbol of the country's manufacturing ambitions, the next battle is being fought deeper inside the value chain.

The shift is already visible. Electronics production has grown nearly seven-fold over the past decade, while exports have expanded more than 11-fold, according to EY Partner Saurabh Agarwal. Mobile phones have led the charge, with more than 99% of domestic demand now being met through local manufacturing. But the bigger question is whether India can build the ecosystem around those.

"The middle ground, which was earlier not fully evolved, is now getting set up," says Pankaj Mohindroo, Chairman of the India Cellular & Electronics Association (ICEA). "The entire sub-assembly and component supply chain is getting established."

That middle layer could prove critical to India's ambitions of becoming an electronics hub. The government's Electronics Components Manufacturing Scheme (ECMS), whose outlay was ₹40,000 crore in the FY27 Budget, is aimed precisely at deepening this part of the supply chain. Industry response has been

strong, with investment commitments significantly exceeding the scheme's initial targets.

The opportunity extends well beyond smartphones. Mohindroo points to IT hardware such as servers, laptops, tablets, consumer electronics, air conditioners, wearables, lighting, auto electronics, and strategic and defence electronics as segments where manufacturing is beginning to gain traction.

India is also trying to build the

foundation underneath this manufacturing base. Semicon 2.0, approved in July with a ₹1.27-lakh crore outlay, broadens the semiconductor push towards equipment, materials, design, IP, and supply-chain capabilities.

For Agarwal, the transition is ultimately about areas where value is created. The transformation has been helped by one company more than most—Apple. Its India strategy has not only expanded iPhone manufacturing but has also brought a large supplier ecosystem into the country. More than 40 suppliers and ecosystem partners have established or expanded operations.

Mohindroo sees Apple's entry as an inflection point, particularly after the initial wave of mobile-phone assembly between 2015 and 2019. As the domestic market matured and geopolitical tensions disrupted existing supply chains, Apple's suppliers—including Foxconn, Pegatron and Wistron, followed by Tata Electronics' expansion and acquisition—accelerated the development of manufacturing capacity and the supporting ecosystem.

That ecosystem is beginning to cluster geographically, with NCR/Noida, Bengaluru, Tirupati, and Sriperumbudur emerging as important manufacturing centres.

TAKING SHAPE

The next phase for India's electronics manufacturing is about building the "middle" of the value chain

The Electronics Components Manufacturing Scheme is accelerating investments in components and sub-assemblies,

Apple's manufacturing expansion has attracted suppliers, capital and manufacturing capabilities

India is increasingly competitive with Vietnam and could match or surpass it by 2030, but China has significant advantages